

FILTER DESIGN TEST CASE

Band-Pass Filter - Band 20 (800 MHz) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P4 - Low

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Band-Pass Filter filter targeting Band 20 (800 MHz) for GPS/GNSS Anti-Jamming Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 2757.5 MHz and bandwidth of 357.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Band-Pass Filter
- Frequency Band: Band 20 (800 MHz)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: 128° YX LiNbO3
- Package: CSP (Chip Scale Package)
- Design Tool: Cadence AWR Microwave Office
- Design Center: Irvine Design Center

This specification applies to all variants of the Band-Pass Filter design targeting Band 20 (800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2757.5 MHz
Bandwidth (3 dB):	357.7 MHz
Insertion Loss Target:	<= 1.3 dB
Return Loss Target:	>= 22.0 dB
Out-of-Band Rejection:	>= 37.0 dB
Isolation (Duplexer):	>= 34.0 dB
Q Factor:	1417
Temperature Coefficient:	-25.6 ppm/°C
Die Size:	2.9 x 0.5 mm
Wafer Size:	8-inch
Number of Resonators:	5
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

- 1. Open Cadence AWR Microwave Office and load the Band-Pass Filter template project.
- 2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
- 3. Set design targets: center freq 2757.5 MHz, BW 357.7 MHz.
- 4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
- 5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
- 6. Verify coupling coefficients between resonator stages.
- 7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
- 8. Export GDS-II layout for mask fabrication.
- 9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
- 10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
- 11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 8272 MHz.
- 12. Compare measured results against simulation and specification limits.
- 13. Perform temperature cycling (-40°C to +85°C) and re-measure.
- 14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 537 MHz offset >= 37.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 8.8 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-08-13
Devices Tested: 67
Insertion Loss (meas): 0.99 dB
Return Loss (meas): 24.4 dB
Rejection (meas): 40.0 dB
Isolation (meas): 38.7 dB
Q Factor (meas): 1417
Group Delay Variation: 6.8 ns
Power Handling: +36 dBm
Die Yield: 94.6%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Band-Pass Filter design demonstrated below-target performance for Band 20 (800 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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